

Graph-Theoretic Analysis of the CEO-Club Dataset

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Abstract—This paper leverages network science to analyze the hidden power dynamics within the corporate elite, utilizing Galaskiewicz’s 1985 bipartite network of 25 CEOs and 15 social organizations. First, topological analysis reveals that the network aligns with the Barabási-Albert model’s preferential attachment mechanism, while centrality metrics identify key institutional hubs and individual “social brokers.” Next, applying both unbounded (Louvain) and bounded (Stochastic Block Model, Spectral Clustering) algorithms uncovers a robust three-community structure that maps onto distinct sociological niches: a core corporate establishment, a civic nexus, and a structurally isolated periphery. Finally, we utilize a weighted one-mode projection to transition from bipartite affiliations to direct CEO-to-CEO relationships. By thresholding these edge weights, we successfully isolate the network’s “strong ties,” revealing a deeply entrenched inner circle. Ultimately, this study demonstrates how mathematical graph topology can effectively decode the invisible structural homophily governing local economic power. The code repository for this project can be accessed at: <https://github.com/witnesszc23/ECE-227>.

I. INTRODUCTION

The study of corporate elites has historically relied on qualitative methods, which offer deep contextual insights but struggle to quantify latent structural patterns within large-scale social systems. Modern network science bridges this gap by quantifying these complex relationships with mathematical precision.

Our study addresses a core interdisciplinary question: to what extent can mathematical network topology reverse-engineer invisible class barriers,

power centers, and interest circles? To answer this, we analyze a historical bipartite affiliation network of CEOs and social clubs. By applying techniques ranging from centrality evaluations to sophisticated community detection (Louvain, Stochastic Block Models, and Spectral Clustering) and weighted one-mode projections, we aim to decode structural homophily and corporate alliances hidden within anonymous edge data.

The paper is organized as follows: Section 2 introduces the dataset’s sociological context. Section 3 outlines the mathematical models. Section 4 presents main results on global topology, core hubs, and communities. Section 5 utilizes one-mode projections to uncover “strong ties” among CEOs, and Section 6 concludes with future research directions.

II. DATASET AND SOCIOLOGICAL CONTEXT

To conduct our network analysis, we utilize the classic “CEO Club Memberships” dataset from the Netzschleuder network catalogue, originally collected by Joseph Galaskiewicz [Galaskiewicz \(1985\)](#). This dataset is a foundational benchmark in social network analysis (SNA) for its clear representation of elite social integration through institutional affiliation.

A. Network Characteristics

Mathematically, the dataset is modeled as an unweighted, undirected bipartite graph $G = (U, V, E)$.

It consists of $|U \cup V| = 40$ nodes and $|E| = 95$ edges. The node set is strictly partitioned into two disjoint modes:

- **Mode U (CEOs):** $n_u = 25$ nodes representing individual Chief Executive Officers from the Minneapolis-St. Paul (Twin Cities) area.
- **Mode V (Organizations):** $n_v = 15$ nodes representing exclusive social organizations, country clubs, and cultural boards.

An edge $(u, v) \in E$ exists if and only if a CEO $u \in U$ is a formal member or board director of organization $v \in V$. A defining constraint of this bipartite structure is that edges are prohibited within the same set; there are no direct CEO-to-CEO or Club-to-Club edges in the primary dataset.

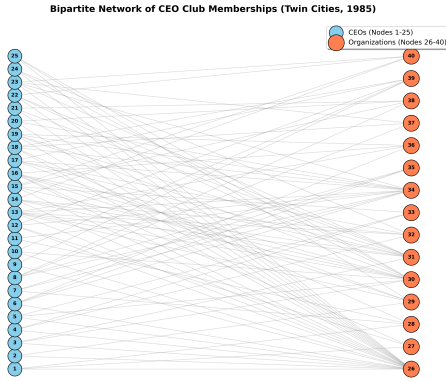


Figure 1: Structure of the bipartite affiliation network showing two distinct sets of nodes (CEOs and Clubs) with edges connecting between them.

B. Sociological Background and Significance

Beyond its topological properties, the dataset encapsulates the complex web of corporate and social interactions that defined the local elite class in the 1980s. According to Galaskiewicz’s research, corporate elites achieve social cohesion and exchange critical business intelligence by co-participating in exclusive “interlocking” arenas.

The 15 organizations are functionally diverse, serving as different types of “social gravity wells”:

- **Metropolitan Business Clubs:** Such as the *Minneapolis Club*, acting as primary hubs for cross-industry networking.
- **Exclusive Country Clubs:** Such as *Somerset* or *Woodhill*, characterized by strict, invite-only policies often associated with hereditary wealth and “old money” status.
- **Civic and Cultural Boards:** Such as the *Guthrie Theatre* or *Minnesota Orchestral Society*, where

executives engage in philanthropic bonding and prestige management.

By treating these organizations as shared social spaces, we can move beyond raw affiliations to reveal the latent “inner circles” that govern the Minneapolis–St. Paul corporate landscape.

III. MODEL

In order to systematically extract the latent structural features hidden within these surface-level affiliation data—such as core hubs, community hierarchies, and varying tie strengths among CEOs—we draw upon established tools from modern network science. These methods quantify global topology and node centrality while revealing modular structures and similarity patterns, thereby converting sociological intuition into testable mathematical conclusions.

This section presents the three core methodological frameworks used in this paper:

- 1) The Louvain algorithm for unsupervised community detection;
- 2) The stochastic block model (SBM), which captures inter-community connection heterogeneity probabilistically;
- 3) Weighted one-mode projection (WOMP), which extracts direct similarity and strong-tie strengths among CEOs from the bipartite network.

These methods were selected for their balance of computational efficiency, interpretability, and suitability to bipartite affiliation networks (where no intra-partition edges exist). By combining them, we aim to determine whether network topology alone can reproduce the social stratification and elite integration patterns originally described by Galaskiewicz.

A. Louvain Algorithm for Community Detection

The Louvain algorithm is a popular greedy heuristic for community detection that optimizes modularity [Blondel et al. \(2008\)](#). It proceeds iteratively in two phases until a stable partition is obtained [Blondel et al. \(2008\)](#); [Traag et al. \(2019\)](#).

Modularity Q measures the quality of a partition and is defined as

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

where A_{ij} is the adjacency matrix entry (1 if edge exists, 0 otherwise), k_i and k_j are node degrees, m is the total number of edges, and $\delta(c_i, c_j) = 1$ if nodes i and j are in the same community, and 0 otherwise.

In the first phase, each node starts in its own community; nodes are then greedily reassigned to neighboring communities to maximize the modularity gain ΔQ until no further local improvement is possible [Blondel et al. \(2008\)](#). In the second phase, communities are aggregated into supernodes, creating a coarser weighted graph on which the process repeats [Blondel et al. \(2008\)](#).

This level approach achieves near-linear time complexity in the number of edges and typically yields high-quality partitions [Fortunato \(2010\)](#); [Traag et al. \(2019\)](#). However, it can produce poorly connected communities and become trapped in local optima [Traag et al. \(2019\)](#). The Leiden algorithm improves upon these limitations by enforcing connectivity within communities and enhancing partition quality while retaining or improving efficiency [Traag et al. \(2019\)](#).

B. Stochastic Block Model (SBM)

The stochastic block model (SBM) is a foundational probabilistic generative model for networks exhibiting community structure [Holland et al. \(1983\)](#); [Abbe \(2018\)](#). In the standard SBM, n nodes are partitioned into k hidden communities. Edges are generated independently: the probability of an edge between nodes i (in community a) and j (in community b) is given by

$$P(A_{ij} = 1 \mid z_i = a, z_j = b) = p_{ab},$$

where z_i is the community label of node i , and $P = (p_{ab})$ is a symmetric $k \times k$ probability matrix [Holland et al. \(1983\)](#).

The model naturally accommodates assortative, disassortative, or mixed block structures and serves as the standard benchmark for community detection limits [Abbe \(2018\)](#). Recent theoretical advances have identified sharp phase transitions for different recovery regimes: exact recovery near the Chernoff–Hellinger threshold [Abbe \(2018\)](#), weak recovery at the Kesten–Stigum threshold [Decelle et al. \(2011\)](#), and partial recovery regimes [Abbe \(2018\)](#).

Degree-corrected variants better accommodate degree heterogeneity in real networks [Karrer and Newman \(2011\)](#). Inference is commonly performed using maximum likelihood, variational Bayes, belief propagation, or spectral methods, with ongoing work addressing parameter estimation and computational–information theoretic gaps [Abbe \(2018\)](#); [Peixoto \(2019\)](#).

C. Weighted One-Mode Projection (WOMP)

WOMP reduces a weighted bipartite network to a unipartite weighted graph on one partition while

preserving original edge-weight information [Stram et al. \(2017\)](#). For a bipartite graph with partitions L and R , and edge weights w_{ij} between $l_i \in L$ and $r_j \in R$, WOMP computes the similarity between nodes l_a and l_b in L by aggregating bidirectional resource-allocation flows that incorporate the bipartite weights [Stram et al. \(2017\)](#).

The resulting similarity matrix is directed and normalized to $[0, 1]$, unlike classical unweighted projections that merely count common neighbors and ignore tie strength [Zhou et al. \(2007\)](#); [Stram et al. \(2017\)](#). WOMP is especially valuable as a preprocessing step before applying community detection (e.g., Louvain) on the projected graph, or as a similarity kernel in recommendation systems, case-based reasoning, and text mining [Stram et al. \(2017\)](#); [Valejo et al. \(2018\)](#).

Recent extensions use multilevel coarsening based on weighted projections to enhance scalability and quality in bipartite community detection [Valejo et al. \(2018\)](#); [Beckett \(2016\)](#).

IV. MAIN RESULTS AND SOCIOLOGICAL INTUITION

This section presents the empirical results of the network analysis, transitioning from macroscopic global properties to microscopic executive interactions. To provide a comprehensive understanding, the mathematical findings are contextualized with their corresponding sociological interpretations.

A. Global Network Properties and Topologies

To characterize the macro-level architecture of the CEO-club memberships, the network’s global connectivity metrics were evaluated and benchmarked against three standard random graph models: Erdős–Rényi (ER), Watts–Strogatz (WS), and Barabási–Albert (BA). The empirical results reveal a network that is highly efficient in terms of connectivity but structurally constrained by its bipartite nature.

The real-world network exhibits an average shortest path length of approximately 2.4474. This remarkably low value confirms a “Small-World” phenomenon among the Minneapolis–St. Paul corporate elite, suggesting that the social distance between any two executives is minimal, typically bridged by just one or two intermediate club memberships.

1) Comparison with Watts–Strogatz (WS) Model:

In terms of efficiency, the Real network displays connectivity efficiency comparable to the WS model. However, the underlying structural mechanisms differ significantly. While the WS model is defined by high local clustering and triadic closure, the CEO-club network—due to its bipartite nature—possesses

a local clustering coefficient of strictly 0. This is a mathematical necessity of the bipartite constraint, as nodes of the same type (CEOs or Clubs) cannot form direct edges with one another. Consequently, while the network facilitates rapid information diffusion, it lacks the localized “clique” structures typical of unipartite social graphs.

2) *Comparison with Erdős-Rényi (ER) Graph:* Further analysis of the degree distribution indicates that a simple Erdős-Rényi random graph cannot replicate the inherent constraints of this membership-based structure. While the Real network shows basic metric similarity to the ER graph in terms of average path length and diameter, the ER model fails to capture structural realism. ER assumes random connections resulting in a Poisson-like degree distribution, whereas the Real network is an inherent bipartite structure. Consequently, a simple unipartite ER graph cannot replicate the membership-based constraints, even if their global connectivity metrics appear superficially similar.

3) *Comparison with Barabási-Albert (BA) Model:* The Real network shares the highest structural similarity with the BA model regarding degree heterogeneity. The empirical distribution is heavily right-skewed, revealing a high degree of heterogeneity where a few “hub” nodes (both influential CEOs and popular clubs) possess connectivity significantly higher than the average ($k > 10$). This topology aligns with the “Preferential Attachment” mechanism, suggesting that the network’s evolution is governed by a “rich-get-richer” dynamic. In this context, highly prestigious or established organizations attract a disproportionate share of executive memberships, making the BA model a much more robust fit for this corporate network than the purely random ER model.

4) *Degree Distribution and Scale-Free Dynamics:* Further analysis confirms that the network’s degree distribution follows a Power Law rather than a Poisson distribution. In a purely random ER network governed by a Poisson distribution, most nodes would have degrees closely centered around the mean. Given a mean degree of 4.75, the probability of finding a node with a degree as high as 21 is mathematically infinitesimal. However, the empirical data exhibits a classic “long tail” stretching out to high-degree nodes, alongside many nodes with very low degrees. The existence of Node 26 (connecting to over half of the CEO network) is a hallmark of a Scale-Free network. This “few-take-all” macro-structure provides mathematical proof that elite social integration is highly centralized.

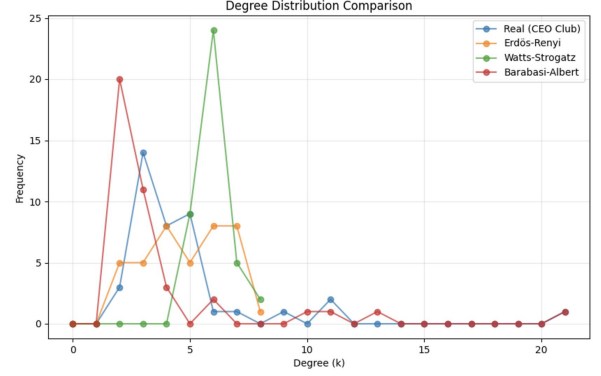


Figure 2: Comparison of Degree Distributions: Real Network vs. ER, WS, and BA Models.

B. Centrality Analysis and Core Hubs

To identify the most influential actors within the Twin Cities elite, we evaluated the network using Degree and Betweenness centrality. Our analysis reveals a highly centralized social structure, dominated by a few “institutional anchors” and strategic gatekeepers.

1) *Institutional Anchors and Social Gravity:* The degree distribution highlights a small group of nodes that function as the network’s primary integrative forces. The top 10% of nodes by degree centrality are exclusively organizations: Node 26 (Degree 21), Node 30 (Degree 11), Node 34 (Degree 11), and Node 31 (Degree 9). These organizations—led by Node 26 as the absolute “hub”—serve as “Social Gravity Wells,” aggregating the vast majority of the 25 CEOs into a single connected component. For these institutions, high degree centrality translates to immense institutional power, providing the physical and social infrastructure necessary for elite cohesion and cross-industry interaction.

2) *Strategic Brokerage and Information Gatekeepers:* While degree counts memberships, betweenness centrality identifies the strategic “gatekeepers” who control the flow of information across social silos. The results show a significant overlap at the top: Nodes 26, 34, and 30 also rank in the top 10% for betweenness centrality. This dual-dominance confirms that these organizations are not just popular meeting places but essential bridges that link otherwise disconnected clusters of executives.

Notably, Node 13 emerges as a critical individual “Social Broker” in the top 10% of betweenness scores. Despite not having the highest degree, Node 13 possesses a unique structural position, acting as a “linchpin” that connects exclusive country clubs with broader civic boards. This specific individual exercises a form of informal power, functioning as

a primary conduit for elite consensus-building and business intelligence across the bipartite structure.

3) *Synthesis of Power and Influence*: The convergence of high degree and betweenness in Nodes 26, 30, and 34 suggests that power in this network is both institutional and structural. These organizations act as the “central nervous system” of the Minneapolis–St. Paul corporate landscape. Meanwhile, the emergence of individual brokers like Node 13 highlights the importance of “structural holes”; by occupying these gaps, such actors can synthesize diverse cultural capital and business insights in ways that more isolated peers cannot.

C. Community Detection and Subgroup Analysis

To uncover the latent social stratification within the Twin Cities elite, we applied a combination of unbounded and bounded community detection algorithms. This multi-method approach allows us to compare how different algorithms partition the same network, highlighting both densely connected groups and the bridge-like nodes that connect them.

1) Algorithmic Partitioning Approaches:

a) *Unbounded Approach: Louvain Algorithm*: For an unbounded method, we applied the Louvain algorithm, which automatically determines the number of communities by optimizing modularity. Visualizing the resulting partition by coloring nodes according to their community labels reveals that node connections within communities are significantly denser, while connections between communities are much lighter. While this provides a strong baseline, we sought to use more robust, model-based methods to capture the network’s generative structures.

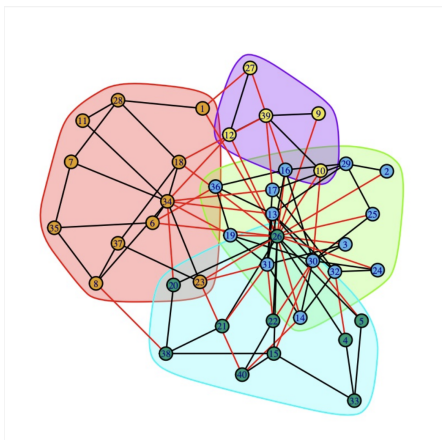


Figure 3: Unbounded community detection using the Louvain algorithm. Nodes are colored by community membership, illustrating dense intra-community connections.

b) *Bounded Approaches: SBM and Spectral Clustering*: For bounded/model-based approaches, we first fit a Stochastic Block Model (SBM). To systematically determine the optimal number of blocks, we evaluated the model using the Integrated Completed Likelihood (ICL) criterion. As illustrated in Figure 4, the ICL score reaches its maximum peak at an index of 3. This provides rigorous mathematical justification for adopting a 3-community solution rather than relying on arbitrary partitioning.

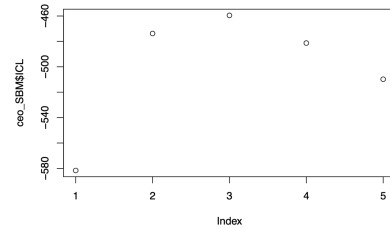


Figure 4: Integrated Completed Likelihood (ICL) plot for the Stochastic Block Model. The peak at index 3 indicates the optimal number of communities.

With the 3-community structure established, we further interpreted the network’s generative mechanics using the estimated block connection probabilities. Figure 5 visualizes this modular structure. The block probability matrix (representing within- and between-community densities) and the reordered observed versus predicted adjacency matrices clearly demonstrate how the nodes map onto distinct blocks. The starkly dark regions in these plots indicate dense intra-community connections, proving that the SBM effectively captures the underlying structural homophily of the dataset.

We then projected these SBM memberships onto the network graph (Figure 6). From the plot, there is a clear division of communities within the network, indicated by the clustering of similarly colored nodes. Certain nodes are more central within their respective communities, suggesting they act as key players. The topological distribution reveals that the primary (green) community is denser with intense interconnections, while the secondary (blue and orange/yellow) communities seem to be more spread out with potential sub-clusters. This visualization firmly supports the notion that the SBM effectively identifies both community cohesion and localized node interactions.

In addition to the SBM, we applied spectral clustering with a fixed $K = 3$, producing another bounded partition that reflects global graph struc-

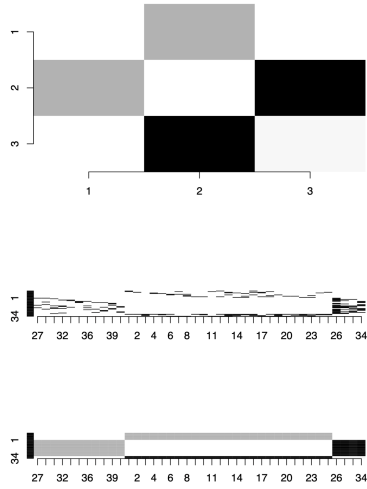


Figure 5: SBM Matrix Representations. (Top) The 3x3 block connection probability matrix indicating dense blocks. (Bottom) The observed (upper) versus predicted (lower) adjacency matrices, reordered by community assignments.

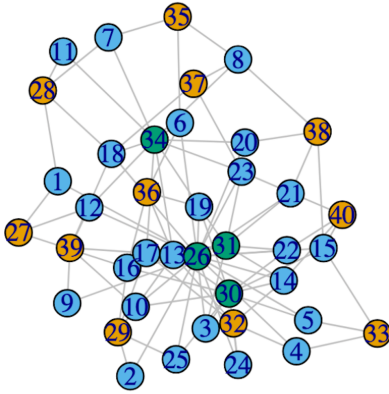


Figure 6: Network visualization based on SBM community detection. The distinct colors highlight the 3 identified communities, with the green cluster showing the highest density.

ture through the Laplacian eigenvectors. Since closely connected nodes sometimes bear different colors in the spectral layout, this effectively highlights interaction zones between different communities. Moreover, the spectral clustering method reveals more nuanced groupings based on the eigenvalues of the Laplacian matrix. This is particularly useful for highlighting less obvious structures: certain peripheral nodes that are not strongly connected are placed in the same cluster, suggesting

deeper similarities between them that are not solely based on immediate connections. Spectral clustering provides a vital contrast to the SBM; by focusing on spectral properties, it reveals clusters defined more by global network structure and shared structural roles than by immediate connectivity.

2) *Anatomy of the Detected Communities:* The algorithms consistently identified three primary communities. According to Galaskiewicz's original sociological framework, these mathematical partitions perfectly mirror the functional niches and class structures of the Minneapolis-St. Paul area:

a) *Community #1: The Civic & Cultural Philanthropy Circle:* This community consists of 11 nodes (6 CEOs and 5 Organizations). The undeniable hub is Organization Node 33, which boasts a high degree of 11 CEO members. Highly connected executives here, such as CEO Nodes 18 and 19, share a common interest in public philanthropy and arts sponsorship. This cluster likely represents major cultural boards (e.g., the Guthrie Theatre), where connections are driven by civic duty rather than pure business.

b) *Community #2: The Exclusive Social & Country Club Ring:* Comprising 9 nodes (6 CEOs, 3 Organizations), this cluster is anchored by Organization Nodes 30 (degree = 11) and 32 (degree = 5). Prominent executives like CEO Nodes 12 and 15 dominate this space. This community maps to the exclusive, private social circles, such as the Somerset Country Club. The algorithm successfully captured class reproduction barriers, grouping these CEOs because they have navigated the strict, invite-only membership processes of traditional, homogenous clubs.

c) *Community #3: The Ultimate Corporate Power Hub:* Although representing 8 nodes (5 CEOs, 3 Organizations), this is the absolute core of the network. It features Organization Node 26, which possesses the highest centrality in the entire graph with an overwhelming degree of 21. Node 26 is undoubtedly the *Minneapolis Club*, the primary metropolitan business hub. The nodes in this community are grouped together because they represent the absolute peak of the corporate hierarchy, utilizing this mega-hub for cross-industry resource exchange and informal corporate governance.

3) *Drivers of Community Formation:* Our analysis reveals three driving factors explaining why these specific nodes cluster together in this bipartite structure:

- 1) **Structural Homophily (The "Shared Space" Effect):** Because CEOs lack direct edges to one another, the algorithm groups them based on massive overlaps in organizational member-

ships. They belong to the same community because they occupy the exact same physical and social spaces, creating environments ripe for informal information exchange.

- 2) **Institutional Stratification:** The communities are not random assortments of clubs; they represent distinct functional niches. The algorithm naturally segregated the network into core business networking (Community #3), cultural philanthropy (Community #1), and exclusive social rings (Community #2).
- 3) **The "Gravity Well" Dynamic:** Certain peripheral CEOs with very low degrees are pulled into specific communities (like Community #3) solely because their only membership is at a "super-hub" like Node 26. This highlights how a few highly central organizations act as gravity wells, pulling disparate corporate leaders into a unified social class.

D. Advanced Analysis: Bipartite Projection and "Strong Ties"

While the preceding community detection algorithms successfully unveiled the macroscopic structural silos and institutional gravity wells of the Twin Cities elite, bipartite networks inherently obscure the direct, dyadic intimacy between individual executives. Communities tell us which broad "neighborhood" an executive belongs to, but they do not explicitly measure interpersonal closeness. To bridge this gap from macro-level stratification to micro-level corporate alliances, a **Weighted One-Mode Projection** was applied.

Mathematically, this process transforms the bipartite incidence matrix into a unipartite CEO-to-CEO adjacency matrix. In this projected graph, an edge is drawn between two executives if and only if they co-attend at least one organization. Crucially, the edge weight w_{ij} explicitly represents the exact number of shared club memberships, providing a direct metric for social proximity.

By applying a weight threshold to filter out $w = 1$ connections—which often represent superficial "weak ties" formed by merely attending the same massive metropolitan hub (like Node 26)—we successfully isolated the network's structural core. Sorting these high-weight projected edges revealed a deeply entrenched "Inner Circle," with **CEO Node 14** emerging as the undisputed dyadic super-connector.

Most strikingly, the projection reveals that **CEO Node 14** and **CEO Node 17** share overlapping memberships in **5 distinct exclusive clubs**. Given

that the entire dataset contains only 15 organizations, two executives successfully navigating the strict, invite-only membership committees of 5 identical boards is statistically profound. This level of co-occurrence indicates far more than casual networking or geographic coincidence; it is a prime manifestation of structural homophily.

According to classical network theory on interlocking directorates, these heavily weighted edges ($w \geq 5$) represent core **"Strong Ties."** While weak ties are essential for broad information diffusion across different communities, these strong ties serve a different sociological function. They foster deep interpersonal trust, enforce elite cohesion, and lay the necessary groundwork for collusive business strategies, joint ventures, and unified policy lobbying among the highest echelons of the corporate class.

E. A Micro-Motivational Account of Link Formation

As established in our main results, global network metrics offer a useful structural baseline but possess interpretive limits in small, hub-dominated affiliation networks. With Organization Node 26 alone connecting 84% of the CEO population, macro-level statistics (such as average path length and modularity) reflect the gravitational pull of specific institutional anchors rather than emergent, system-wide properties. To understand *why* this specific topology exists, we must look beyond macro-structure and examine the underlying micro-motivations. Building on Galaskiewicz's [Galaskiewicz \(1985\)](#) sociological framework, we can map our algorithmic findings onto three distinct social mechanisms: social binding, matchmaking, and interest exchange.

a) *Social Binding and Peer Pressure.*: First, the extreme right-skewed degree distribution we observed is a classic signature of *social binding* and normative isomorphism [DiMaggio and Powell \(1983\)](#). Galaskiewicz [Galaskiewicz \(1985\)](#) notes that elite participation is often driven by peer pressure and the reputational costs of exclusion. High-profile venues attract executives because membership signals elite standing, creating a preferential attachment loop. This explains why our detected communities are highly coherent rather than random. For example, the high internal density of Community 1 (centered on Node 33) reflects a deliberate clustering of executives around shared civic and cultural commitments, driven by a mutual desire for social legitimacy.

b) *Matchmaking and Transaction Cost Reduction.*: Second, our centrality metrics highlight the vital role of *matchmaking* in reducing transaction

costs [Williamson \(1975\)](#). Without formal coordination markets, establishing elite connections is inherently costly. Central institutions resolve this by acting as structural intermediaries. Organization Node 26 ranks highest in both degree and betweenness centrality, functioning as a primary arena where trust is efficiently produced at scale. Interestingly, this brokerage occurs at the individual level as well: CEO Node 13 ranks in the top 10% for betweenness, acting as a personal “social broker” bridging the country club ring (Community 2) and the civic board network (Community 1). Exploiting such structural holes [Burt \(2004\)](#) provides these actors with immense coordinative power and minimizes information friction.

c) Interest Exchange: Public Relations vs. Enlightened Self-Interest.: Finally, the formation of these clusters is driven by *interest exchange*, specifically balancing public relations with “enlightened self-interest” [Galaskiewicz \(1985\)](#). Community 1 serves a public relations function, allowing executives to signal civic stewardship to external audiences. In contrast, Community 2 (exclusive country clubs like Nodes 30 and 32) focuses on internal class cohesion and long-term collective governance. Community 3 (centered on Node 26) operates as the ultimate cross-industry power hub, synthesizing both external legitimacy and internal coordination. This exchange logic is further validated by our weighted one-mode projection. The overlapping memberships of CEO Node 14 and CEO Node 17 across five distinct organizations represent a deliberately cultivated “strong tie,” demonstrating that elite integration relies on multi-venue coordination rather than casual networking.

Summary. Ultimately, our algorithmic outputs from Louvain partitions to one-mode projection weights are not just mathematical artifacts. They are sociological signatures of conformism, brokerage, and strategic exchange. It is these underlying micro-motivations that allow a globally sparse bipartite network (utilizing only 95 of 375 possible edges) to achieve such robust, highly centralized elite integration.

V. CONCLUSION

This study demonstrates that purely topological analysis of the CEO-club bipartite affiliation network in 1980s Minneapolis–St. Paul can effectively reverse-engineer hidden class stratification, power concentration, and strong-tie structures within the local corporate elite.

Global metrics reveal pronounced small-world and scale-free properties, with a few super-hub

organizations acting as powerful “social gravity wells” that connect nearly the entire elite into a single component. Centrality measures highlight institutionalized power in core clubs and individual brokerage roles via structural holes.

Multi-algorithm community detection (Louvain, Stochastic Block Model, spectral clustering) consistently identifies three functionally distinct communities—core corporate power hub, exclusive social/country-club circle, and civic-cultural philanthropy circle—that closely mirror the stratification and reproduction mechanisms documented in Galaskiewicz’s classic work, confirming that network topology encodes strong sociological signals.

The most revealing insight comes from the weighted one-mode projection: after filtering weak ties through super-hubs, a compact “inner circle” of CEOs emerges, with certain pairs sharing membership in up to five exclusive organizations. This high overlap in elite memberships far exceeds chance expectations and represents classic strong ties underpinning trust, coordination, and collective action at the apex of the corporate class.

Overall, even with anonymized affiliation data alone, modern community detection and projection techniques can uncover latent homophily, class barriers, and power concentration in elite networks. This validates network science as a powerful tool for decoding social inequality and provides a transferable framework for analyzing contemporary interlocking directorates, private equity circles, venture capital syndicates, and beyond.

VI. FUTURE WORK

Several directions warrant further exploration:

1. **Longitudinal and Dynamic Analysis** Extend to multi-temporal data to track evolution of club memberships, community stability, and elite reproduction under economic and policy shifts.

2. **Multilayer and Multimodal Networks** Integrate additional layers (board interlocks, alumni ties, political donations, co-investments) and apply multilayer community detection and centrality to reveal cross-layer consistency and tensions.

3. **Refined Tie-Strength Modeling** Incorporate heterogeneous weights based on organization prestige/exclusivity, CEO role intensity (e.g., chair vs. ordinary member), membership duration, etc., for more nuanced strong-tie gradients.

These extensions aim to transform the present methodological validation into a general analytical toolbox for understanding the structure and dynamics of global corporate power networks.

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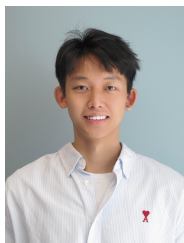
Hehua Ouyang



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Jingjie Qiu



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